Love of Variety and Dynamic Advertising

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Roadmap

1. Motivation
2. Model
3. Simulation Design
4. Results: Consumer Behavior
5. Next Steps

Motivation RETURN TO THIS

- Amazon anti-trust scrutiny
  - Amazon accounts for almost 40% of US e-commerce sales (Statista, 2023)
  - Amazon's dual role as retailer and platform raises concerns about anti-competitive behavior
  - Can use own data and competitors' data to more effectively target ads and product introductions
- Consumer agency over data (opt out)
  - Apple iOS 14.5 update: 85% opt-out rate upon rollout (Flurry Analytics, 2022)
  - Facebook ad revenue down $13bn in 2022 due to app tracking transparency (ATT) policy

Key Concern

Data obfuscation etc.

Why Does Variety Matter?

Research Question

How do consumer preferences for variety impact the firms' targeting strategy and what are the welfare implications of data privacy policies that limit ad targeting?

- Understanding the role of variety-seeking is valuable for modeling consumer behavior and evaluating the impact its accounting has on estimates of consumer welfare.
- Canonical logit models ignore consumer preferences for variety, which are potentially empirically important
- These preferences may shape consumer switching behavior, which in turn shapes firms' incentives to strategically target ads and introduce new products.

What This Paper Does (So Far)

1. **Model:** Dynamic discrete choice model of consumer behavior with love of variety and strategic advertising by firms
2. **Simulation:** Characterize consumer and firm behavior across preference regimes and advertising specifications
3. **Estimation (coming):** Bring model to Nielsen scanner data to recover structural parameters
4. **Counterfactuals (coming):** Study advertising targeting policy and welfare implications

Today's talk: Consumer demand function + simulation results

Literature Connections

Demand side

- Lattin (1987): State dependence in consumer choice
- Chintagunta (1999): Extension to panel data
- Dubé, Hitsch, Rossi (2009): Habit formation and switching costs

Advertising / firm side

- Shaffer and Zhang (1995): Theoretical basis for ad targeting in a two-period model
- Erdem & Keane (1996): Learning dynamics
- Zhang, Choi, Cheng (2025): Updated targeting game

Contribution

Unite the demand and supply side literatures by modeling consumer preferences for variety in a dynamic discrete choice framework with strategic advertising.

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Consumer's Problem

Consumer i chooses product j in period t to maximize:

$$U_{ijt} = \underbrace{\beta_{ij} X_{jt}}_{\text{characteristic component}} + \underbrace{\gamma_i \log(1 + \Sigma_{ijt}^2)}_{\text{novelty component}} - \underbrace{\alpha\, p_{jt}(1 - a_{jt})}_{\text{ad-adjusted price}} + \xi_j + \varepsilon_{ijt}$$

- where $\Sigma_{ijt} = X_{jt} - \theta_{it}$ is the deviation of product j's characteristic from consumer i's previously chosen characteristics θ_{it} in period t.
- at the moment, $\theta_{it} = \frac{1}{t}\sum_{s=1}^{t-1} X_{js}$, or the mean of previously chosen product characteristics.

Key parameters:

- $\beta_i > 0$: quality
- $\gamma_i > 0$: variety-seeking

Choice probability (MNL):

$$P_{ijt} = \frac{e^{U_{ijt}}}{1 + \sum_k e^{U_{ikt}}}$$

Follows from $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$

State Variable: History of Choices

History of Choices θ_{it}

$$\theta_{it} = f(\{X_{js}\}_{s<t}) = \frac{1}{t}\sum_{s=1}^{t-1} X_{js}$$

The consumer's history is summarized by the mean of previously chosen product characteristics.

Simulation design:

- **Initial state:** Simulate $T_{\text{prior}} = 5$ pre-sample periods, choose products from $J \in [1, 2, 3, 4, 5]$ to generate an experience-based prior.

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Simulation Parameters

Table: Parameters

Param.	Value	Description
T	100	Periods
J	5	Products
S	1,000	Simulations
β_{ij}	2	Quality
γ_i	$\{6, 9, 12\}$	Variety-seeking
α	0	Price (off)
ξ_j	0	Unobserved quality (off)
β_{df}	0	Discount factor (off)

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Utility Measurement (No LOV)

Utility Function without LOV

$$\hat{U}_{ijt} = \beta_{ij} X_{jt} + \epsilon_{ijt}$$

- Any variation due to ε_{ijt}, not consumer preferences
- CCPs are not informative about consumer behavior or firm incentives
- Nearly all mass on Product 5, which has the highest quality X_{jt}

Table: CCPs (No LOV)

Product	CCP
Product 1	0.02%
Product 2	0.84%
Product 3	4.40%
Product 4	19.3%
Product 5	75.3%

Consumer Behavior with LOV

Utility Function with LOV

$$\hat{U}_{ijt} = \beta_{ij} X_{jt} + \gamma_i \log(1 + \Sigma_{ijt}^2) + \epsilon_{ijt}$$

- where $\Sigma_{ijt} = X_{jt} - \theta_{it}$ as before.
- Consumer behavior varies significantly across variety specifications.
- Higher γ_i leads to more switching.
- Cross-sectional sorting: consumers with higher γ_i are more likely to switch to lower indexed products.

LOV Specification

	γ			
	0	6	9	12
Conditional Choice Probabilities				
Product 1	0.0016	0.3313	0.3971	0.4234
Product 2	0.0084	0.0551	0.0081	0.0009
Product 3	0.0437	0.0027	0.0000	0.0000
Product 4	0.1934	0.0023	0.0002	0.0000
Product 5	0.7529	0.6086	0.5946	0.5756
Inclusive Values				
Inclusive Value	10.9606	17.4522	22.4506	27.5139
IV at $t = 50$	—	16.3861	21.5165	27.6856

Product Introduction ($\gamma = 0$)

	Value of Introduced Product (q_6)				
	$q_6 = 0$	$q_6 = 1$	$q_6 = 3$	$q_6 = 5$	$q_6 = 6$
Product 1	0.0016	0.0016	0.0016	0.0012	0.0009
Product 2	0.0089	0.0089	0.0087	0.0066	0.0053
Product 3	0.0440	0.0439	0.0429	0.0332	0.0263
Product 4	0.1927	0.1926	0.1883	0.1484	0.1176
Product 5	0.7527	0.7522	0.7375	0.5927	0.4727
Product 6	0.0001	0.0008	0.0210	0.2179	0.3771
$\%\Delta$ IV	0.9993	0.9994	1.0017	1.0362	1.0910
IV at $t = 50$	11.3297	11.3300	11.3491	12.0579	13.5177

Product Introduction ($\gamma = 6$)

	Value of Introduced Product (q_6)				
	$q_6 = 0$	$q_6 = 1$	$q_6 = 3$	$q_6 = 5$	$q_6 = 6$
Product 1	0.2125	0.2665	0.3295	0.2973	0.2396
Product 2	0.0353	0.0446	0.0571	0.0523	0.0525
Product 3	0.0021	0.0025	0.0031	0.0029	0.0043
Product 4	0.0018	0.0019	0.0021	0.0017	0.0012
Product 5	0.5943	0.5796	0.6069	0.4769	0.3082
Product 6	0.1540	0.1049	0.0013	0.1689	0.3941
$\%\Delta$ IV	1.0126	0.9961	0.9835	0.9975	1.0946
IV at $t = 50$	20.9308	19.6201	16.6054	19.8850	27.0847

Product Introduction ($\gamma = 9$)

	Value of Introduced Product (q_6)				
	$q_6 = 0$	$q_6 = 1$	$q_6 = 3$	$q_6 = 5$	$q_6 = 6$
Product 1	0.2087	0.3131	0.3966	0.3643	0.3164
Product 2	0.0054	0.0072	0.0093	0.0088	0.0122
Product 3	0.0001	0.0001	0.0001	0.0001	0.0001
Product 4	0.0001	0.0001	0.0001	0.0001	0.0001
Product 5	0.5768	0.5616	0.5940	0.4628	0.2994
Product 6	0.2088	0.1179	0.0000	0.1640	0.3719
$\%\Delta$ IV	1.0301	0.9932	0.9823	0.9932	1.0928
IV at $t = 50$	24.0019	21.8304	21.5281	22.0987	30.4871

Product Introduction ($\gamma = 12$)

	Value of Introduced Product (q_6)				
	$q_6 = 0$	$q_6 = 1$	$q_6 = 3$	$q_6 = 5$	$q_6 = 6$
Product 1	0.2113	0.3321	0.4234	0.3913	0.3503
Product 2	0.0007	0.0009	0.0011	0.0011	0.0019
Product 3	0.0000	0.0000	0.0000	0.0000	0.0000
Product 4	0.0000	0.0000	0.0000	0.0000	0.0000
Product 5	0.5546	0.5451	0.5755	0.4494	0.2904
Product 6	0.2333	0.1219	0.0000	0.1582	0.3574
$\%\Delta$ IV	1.0421	0.9908	0.9818	0.9906	1.0930
IV at $t = 50$	32.1484	29.3938	29.3151	29.3385	37.1444

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Road to Estimation

1. **Consumer value function**
   Move from reduced-form simulation to solving the full dynamic discrete choice with continuation values
2. **Nielsen scanner data**
   Bring model to data to recover structural parameters; use variation in consumer switching behavior to identify variety-seeking parameter
3. **Firm's Problem**
   Ad targeting policy; profit maximization; product introduction; welfare implications of data privacy policies

Summary

1. **Model:** Consumer utility depends on deviation from experiences; nests variety-seeking
2. **History:** Rich behavioral dynamics; greater preference for variety generates cross-sectional sorting
3. **Next:** Multiple consumers, structural estimation, Nielsen data

Thank you — questions welcome

Appendix

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Simulation Algorithm (Consumer)

1. Draw $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$ for all i, j, t
2. Initialize $\bar{X}_{i0}$ (set mean or prior mean)
3. For $t = 1, \ldots, T$:
   1. Compute $U_{ijt} = \beta_i X_{jt} + \gamma_i \log(1 + (X_{jt} - \bar{X}_{it})^2)\varepsilon_{ijt}$
   2. Set $j^* = \arg\max_j U_{ijt}$
   3. Update $\bar{X}_{i,t+1}$
4. Compute CCPs via log-sum-exp for numerical stability: $\log P_{ijt} = U_{ijt} - \log \sum_k e^{U_{ikt}}$
5. Average over $S = 1{,}000$ simulation draws